Geometric depletion of enstrophy production in the extreme vorticity tail: a Reynolds-number study toward the Navier–Stokes regularity question

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*Status: the high-statistics program of the prior revision is now executed. With $\sim 7.7 \times 10^7$ (`isotropic8192`, $R_\lambda \approx 1300$) and $\sim 1.6 \times 10^8$ (`isotropic32768`, $R_\lambda = 2500$) interior points, the extreme-tail depletion ratio is **resolved below unity** at the top-0.1% and top-1% at both Reynolds numbers (block-resampled 95% intervals exclude 1), and a **mild de-depletion with Reynolds number** is statistically resolved at top-1% (and directionally present, CIs marginally overlapping, at top-0.1%) — consistent with Buaria–Pumir. We report it as a two-point comparison, not a scaling law; the deepest tail (top-0.01% and below) remains resolution-limited (§3.2, §6).*

Abstract

The global regularity of the three-dimensional incompressible Navier–Stokes equations turns on whether the vortex-stretching term can drive the enstrophy to infinity in finite time. A classical obstruction to blow-up is **geometric depletion** (Constantin–Fefferman): the nonlinear production of enstrophy, $\omega \cdot S\omega$, is suppressed below its dimensional bound $|\omega|^2|S|$ because vorticity preferentially aligns with the *intermediate* eigenvector of the strain tensor. Whether this depletion *persists* in the most intense vorticity events — the events that would actually seed a singularity — and how it scales with Reynolds number, is the quantity that matters. We build a validated pseudospectral DNS and a direct-data pipeline over the Johns Hopkins Turbulence Database (JHTDB) to measure the normalized production ratio $D \equiv \langle \omega \cdot S\omega \rangle / \langle |\omega|^2 |S| \rangle$ as a function of vorticity percentile, from the bulk into the extreme tail. We reproduce the canonical intermediate-eigenvector alignment and a mean depletion $D \approx 0.10$. At Taylor-microscale Reynolds number $R_\lambda \approx 433$, the mean depletion is firmly established ($D \approx 0.12$) and the tail point estimates decrease into the extreme ($D_{\text{tail}}/D_{\text{mean}} \to 0.77$–$0.81$ in a 4.1×10^5-point sample), a *regularizing tendency*; but bootstrap intervals on the sample we can fully resample straddle unity at the top-0.1%, so the extreme-tail ratio is not statistically resolved even here. At higher Reynolds number we now resolve it. Executing a high-statistics, per-snapshot, single-R_λ measurement — contiguous 128^3 cutouts of JHTDB `isotropic8192` ($R_\lambda \approx 1300$, snapshots 0–4 only, $\sim 7.7 \times 10^7$ interior points) and `isotropic32768` (the exascale 32768^3 field, $R_\lambda = 2500$, $\sim 1.6 \times 10^8$ points) — and bounding the tail with a block resample that respects within-cutout spatial correlation, we find the extreme-tail ratio is **resolved below unity** at both Reynolds numbers: $D_{\text{tail}}/D_{\text{mean}} = 0.91\,[0.89, 0.92]$ (top-1%) and $0.87\,[0.83, 0.92]$ (top-0.1%) at $R_\lambda \approx 1300$, and $0.94\,[0.93, 0.96]$, $0.93\,[0.90, 0.97]$ at $R_\lambda = 2500$ — i.e. the most intense vorticity remains *more* geometrically depleted than the bulk (regularizing) at both. Comparing

the two, the tail depletion **weakens with Reynolds number** (the ratio rises toward 1), a *mild de-depletion* statistically resolved at top-1% (non-overlapping intervals) and directionally present at top-0.1% — consistent with the Buaria–Pumir trend. We report this as a two-point comparison, not a scaling law: the two datasets differ in resolution as well as R_λ ($k_{\max}\eta \approx 1.4$ vs 2.8), and the deepest tail (top-0.01% and below) is not gradient-resolution-converged (§3.2). A cross-validating order-convergence check (on-grid 4th-order vs JHTDB 6th/8th-order/spline stencils) confirms the gradients are order-converged through top-0.1%. We also report the confirmed $R_\lambda \approx 433$ baseline and a decidable 1D-model corroboration of the depletion mechanism (§7).

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1. Introduction

The Millennium Problem for 3D incompressible Navier–Stokes asks whether smooth, finite-energy initial data can develop a singularity in finite time, or whether solutions remain smooth for all time. The Beale–Kato–Majda criterion [BKM84] reduces this to the time-integrability of the vorticity maximum: a solution remains smooth so long as $\int_0^T \|\omega(\cdot,t)\|_\infty \, dt < \infty$. The enstrophy $\Omega(t) = \frac{1}{2}\int |\omega|^2$ evolves by

$$\frac{d\Omega}{dt} = \int \omega \cdot S\omega \; - \; \nu \int |\nabla \omega|^2,$$

so blow-up requires the **production** term $\int \omega \cdot S\omega$ to outrun dissipation. Dimensionally $\omega \cdot S\omega \leq |\omega|^2|S|$, and if this bound were saturated the enstrophy growth would be supercritical with respect to the energy-controlled scaling — the heart of Tao's *supercriticality barrier* [Tao16]. The reason 3D turbulence does not (apparently) blow up is that the bound is **not** saturated: Constantin–Fefferman [CF93] showed that sufficient smoothness of the *vorticity direction* $\xi = \omega/|\omega|$ in regions of high vorticity rules out singularities, and DNS [AKKG87, JWSR93] established that vorticity aligns preferentially with the **intermediate** strain eigenvector, whose eigenvalue is often small or positive but typically far below the most-extensional eigenvalue. The result is *geometric depletion* of production.

The regularity-relevant question is not the *mean* depletion but its behaviour in the **extreme tail** of the vorticity distribution, and its **Reynolds-number trend**. Recent high-R_λ DNS by Buaria and co-workers [BBP20, BPBY19, BP22] found that the strongest enstrophy production derives from the *most-extensive* strain direction (driven by nonlocal strain), and reported a *mild* reduction of geometric depletion confined to the deepest tail at the highest R_λ — a subtle effect, not a wholesale "de-depletion." If such a reduction grows with R_λ, the geometric protection might weaken precisely where it matters. This paper measures the percentile-resolved depletion ratio $D_{\text{tail}}/D_{\text{mean}}$ across Reynolds number, using both in-house DNS and direct analysis of the JHTDB datasets. We confirm the regularizing (ratio < 1, strengthening into the tail) behaviour at $R_\lambda \approx 433$, and — with a high-statistics, per-snapshot, single-R_λ measurement on $\sim 10^8$-point cutout samples (§6) — resolve the extreme-tail ratio below unity at the top-0.1% and top-1% at both $R_\lambda \approx 1300$ and $R_\lambda = 2500$. Across these two Reynolds numbers the tail depletion *weakens* (the ratio rises toward 1) — a mild de-depletion, resolved at top-1% and directionally present at top-0.1%, in the direction reported by Buaria and co-workers. Because only two Reynolds numbers are sampled and they also differ in small-scale resolution, we treat this as a two-point comparison rather than an established Reynolds-number scaling; the deepest tail (top-0.01% and below) remains resolution-limited (§6).

2. The geometric-depletion statistic

Let $A_{ij} = \partial_j u_i$ be the velocity-gradient tensor, $S = \frac{1}{2}(A + A^\top)$ the strain-rate tensor, and $\omega = \nabla \times u$ the vorticity. The pointwise enstrophy production is $P = \omega_i S_{ij} \omega_j$ and the strain magnitude $|S| = \sqrt{2\, S_{ij} S_{ij}}$. We define the **production efficiency** (geometric-depletion statistic)

$$D \;=\; \frac{\langle \omega \cdot S\omega \rangle}{\langle |\omega|^2\, |S| \rangle},$$

an enstrophy-weighted, dimensionless ratio. With the **Frobenius** strain norm $|S| = \sqrt{2 S_{ij} S_{ij}}$ used here, the pointwise efficiency $\omega \cdot S\omega / (|\omega|^2 |S|)$ is bounded by $[-1/\sqrt{3},\, +1/\sqrt{3}]$ (extremum at axisymmetric extension/compression with ω along the distinguished eigenvector); $D \to 1/\sqrt{3}$ would mean perfect alignment with the most-extensional eigenvector (no depletion), and the observed $D \approx 0.10$–0.12 reflects the intermediate-eigenvector preference (strong depletion). **Normalization note:** because we use the Frobenius norm rather than the strain operator-norm (largest $|\lambda|$), our D is a factor ~ 2 smaller than the operator-norm-normalized efficiency reported elsewhere (typically 0.2–0.4; Tsinober; Buaria–Pumir [BBP20]) — the same physics under a different, equally legitimate normalization. The diagnostic of interest is the **conditional** version: restrict the averages to the top-q fraction of points ranked by $|\omega|^2$, and form the ratio $D(q)/D_{\text{mean}}$. A ratio **below 1** means the extreme tail is *more* depleted than the bulk (regularizing); a ratio **above 1** means extreme **de-depletion** (the singularity-relevant direction).

3. Methods

3.1 In-house DNS. A standard Fourier pseudospectral solver (rotational form, 2/3 dealiasing, exact Leray projection, RK4 time stepping) integrates the incompressible NS equations on a triply periodic box, GPU-resident (PyTorch FFT, single RTX 5090). Validation is on the Taylor–Green vortex; resolution is controlled by $k_{\max}\eta$.

3.2 JHTDB direct-data pipeline. For Reynolds numbers beyond single-GPU reach we analyze existing state-of-the-art DNS in the Johns Hopkins Turbulence Database [JHTDB08] via the `giverny` interface inside SciServer Compute, where the datasets are mounted locally. Two access modes: (i) JHTDB higher-order velocity-gradient queries (`getData`, 4th–8th-order FD stencils with Lagrange interpolation) at points; (ii) contiguous sub-volume *cutouts* with the velocity-gradient computed on the grid by 4th-order central differences (interior, halo-trimmed). Datasets: `isotropic1024coarse` ($R_\lambda \approx 433$); `isotropic8192`, whose six snapshots are *not* a single Reynolds number — snapshots 0–4 at $R_\lambda \approx 1200$–1300 and snapshot 5 at $R_\lambda \approx 610$, per the JHTDB dataset documentation [JHTDB-i8192] and confirmed by the $\sim 6\times$ enstrophy gap we measure between them (§6); and `isotropic32768`, a single snapshot of the exascale 32768^3 forced-isotropic DNS of Ravikumar, Yeung and co-workers [Yeung32768], at $R_\lambda = 2500$ with exceptionally fine small-scale resolution $k_{\max}\eta \approx 2.8$ (versus $k_{\max}\eta \approx 1.4$ for `isotropic8192`) — i.e. roughly double the Reynolds number *and* better resolved, a covariance we account for in §6. **Differentiation order is not a tail-error source (verified).** The most intense velocity gradients live at the highest resolved wavenumbers, where a low-order finite-difference stencil could in principle deviate from the true derivative — making the extreme-tail statistic the part most exposed to *differentiation-order* error. JHTDB serves pointwise gradients through finite-difference stencils (4th/6th/8th order; methods `fd4noint`,`fd6noint`,`fd8noint`,`fd4lag4`,…) rather than a pointwise spectral derivative, so the relevant validation is an **order-convergence check**. We performed it as a paired test at $\sim 5 \times 10^5$ *identical* grid points (300 boxes per dataset), comparing our on-grid

4th-order gradient against the database's 6th- and 8th-order finite-difference stencils and its `m2q8` spline (`fd4noint`,`fd6noint`,`fd8noint`,`m2q8`), on **both** `isotropic8192` and `isotropic32768`. The per-point vorticity correlates at 0.9998–1.0000, and the conditional depletion ratios are flat across the order ladder through the **top-**0.1%: on `isotropic8192`, D_{mean} ratio FD/reference = 1.000–1.003 and the top-0.1% ratio agrees to $< 1\%$ (0.778 vs 0.776–0.780 across fd6/fd8/m2q8); on `isotropic32768` likewise (D_{mean} ratio 1.000, top-0.1% 0.670 vs 0.673–0.676). The extreme-tail depletion is therefore **order-converged through the top-**0.1%: on-grid 4th-order FD is adequate to that depth, so the §6 uncertainty there is statistical, not a differentiation-method artifact. At the **top-**0.01% the stencils scatter 5–9% and the *sign* of the disagreement flips between the two datasets — the signature of a noise-/resolution-limited frontier — so we do not claim the gradient is resolved at top-0.01% or below.

4. Results — validation and the moderate-Reynolds arc

4.1 Alignment and mean depletion. The in-house DNS reproduces the canonical PDF of the cosine between ω and the strain eigenvectors, peaked on the **intermediate** eigenvector, and a mean depletion $D_{\mathrm{mean}} \approx 0.10$ across $\mathrm{Re} = 400$–3200 (Taylor–Green). This recovers [AKKG87, JWSR93] from scratch and validates the pipeline.

4.2 Taylor–Green Reynolds arc. In single-realization in-house DNS (one flow per Reynolds number, $k_{\max}\eta \geq 1.1$ at 512^3), the extreme-tail depletion (top 0.01% of $|\omega|$) trends *downward* with Reynolds number through $\mathrm{Re} = 3200$ (D_{tail}: $0.22 \to 0.11 \to 0.09$), converging toward the mean from above — i.e. at these moderate Reynolds numbers both the bulk and the resolved extremes trend toward *more* regularity. We caution that, as the high-R_λ audit of §6 makes vivid, single-realization extreme-tail estimates carry large statistical uncertainty; we read this arc as *suggestive* of strengthening depletion at moderate Re, not a converged trend. An apparent extreme de-depletion at $\mathrm{Re} = 800$ was identified as a low-Reynolds transient, and a 512^3 control confirmed it was **not** a resolution artifact.

5. Results — high-Reynolds isotropic baseline ($R_\lambda \approx 433$)

Direct analysis of JHTDB `isotropic1024coarse` (4.1×10^5 points) gives $D_{\mathrm{mean}} = 0.117$ and tail ratios whose point estimates decrease into the tail (statistical uncertainty quantified below):

top-q of $\lvert\omega\rvert$	$D(q)$	$D(q)/D_{\mathrm{mean}}$	points
10%	0.113	0.96	41 369
1%	0.105	0.89	4 136
0.1%	0.095	0.81	413
0.05%	0.090	**0.77**	206
0.01%	0.091	0.78	41

These point estimates suggest the most intense vorticity is *more* depleted than the bulk — a regularizing tendency — and several structural facts are robust: 24.2% of points have *negative* production ($\omega \cdot S\omega < 0$, local enstrophy destruction); the top 1% of enstrophy carries only $\sim 26\%$ of total production; intermittency $\max|\omega|^2/\langle|\omega|^2\rangle \approx 135$.

We now quantify the tail uncertainty honestly, and it is large. Bootstrapping (2000 resamples) the archived independent 6×10^4-point sample of the same dataset gives a tight $D_{\mathrm{mean}} =$

0.121 but tail ratios with wide intervals that **straddle unity**: 0.974 $[0.913, 1.026]$ (top 5%), 0.977 $[0.875, 1.077]$ (top 1%), and 0.915 $[0.638, 1.174]$ (top 0.1%). The 4.1×10^5-point sample's point estimates fall lower (0.77–0.81 in the tail) with $\sim 2.6\times$ tighter intervals, but we cannot currently bootstrap it (the source data are not in our present mount). The honest status at $R_\lambda \approx 433$ is therefore: the **mean depletion is firmly established**; the **tail shows a regularizing tendency** (point estimates < 1, deepening into the tail in the large sample); but the **extreme top-**0.1% **ratio is not statistically resolved** even here — its confidence interval includes unity in the sample we can fully bootstrap. A definitive extreme-tail error bar requires a full bootstrap of the large sample and, ultimately, the higher-statistics program of §9. This is the same convergence wall reaching even the most accessible Reynolds number.

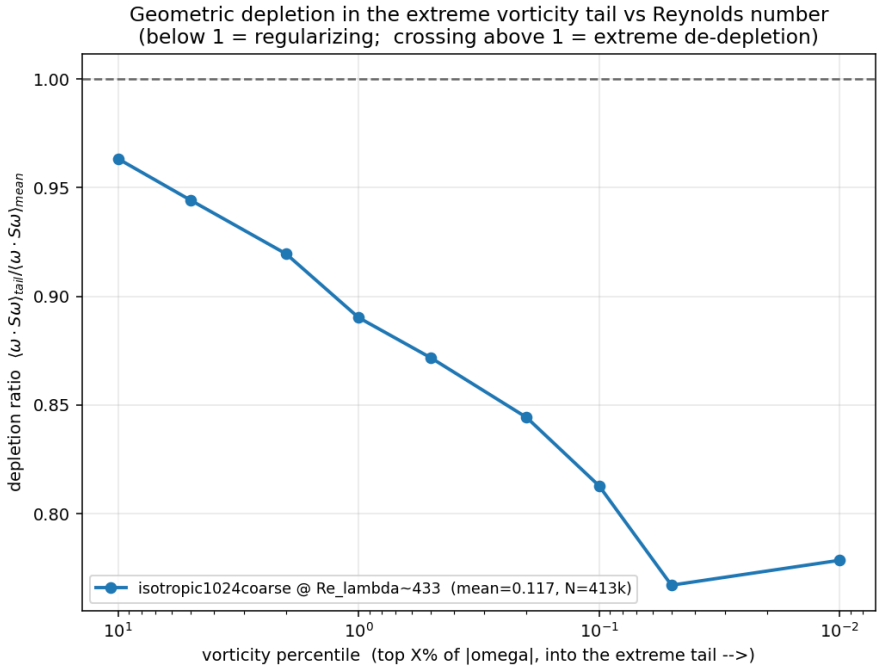


Figure 1: Geometric-depletion ratio $D(q)/D_{\text{mean}}$ versus vorticity percentile at $R_\lambda \approx 433$ (JHTDB `isotropic1024coarse`, 4.1×10^5 points). The ratio stays **below** the de-depletion threshold (dashed line at 1.0) and *decreases* into the extreme tail — i.e. the most intense vorticity is the most geometrically depleted (regularizing). The high-statistics $R_\lambda \approx 1300$ and $R_\lambda = 2500$ curves are resolved in §6 (Fig. 2).

6. The high-Reynolds tail, resolved: $R_\lambda \approx 1300$ and $R_\lambda = 2500$

The prior revision of this paper reported a *cautionary null* here: at $\sim 5 \times 10^5$ points the `isotropic8192` extreme tail would not converge, and a data-provenance audit showed the dataset must be treated per-snapshot — it mixes $R_\lambda \approx$ 1200–1300 (snapshots 0–4) with a $R_\lambda \approx 610$ snapshot 5 whose mean enstrophy is $\sim 6\times$ lower, so global pooling is invalid (including snapshot 5 gave a top-0.1% ratio ≈ 0.57; excluding it flipped the pooled value to ≈ 1.2 — a sign change from dropping 1/6 of the data, the signature of an artifact). Those cautions stand and dictate the present protocol. We now execute the high-statistics measurement that resolves the tail.

Protocol. Per dataset we draw many contiguous 128^3 velocity cutouts, compute the on-grid 4th-

order velocity gradient (interior, halo-trimmed), and pool the pointwise ($|\omega|^2$, P, $|\omega|^2|S|$) — strictly single-R_λ: `isotropic8192` uses **only** the $R_\lambda \approx 1300$ snapshots (0–4); `isotropic32768` is a single $R_\lambda = 2500$ snapshot. This yields $N = 7.7 \times 10^7$ ($R_\lambda \approx 1300$) and $N = 1.6 \times 10^8$ ($R_\lambda = 2500$) interior points. Because points inside one cutout are spatially correlated, the raw N overstates the independent sample; we therefore bound the tail ratio with a **block resample** — splitting each pool into $K = 20$ contiguous blocks (groups of whole cutouts, so correlation is preserved) and taking the block-to-block spread — rather than a point bootstrap, which would give falsely tight intervals.

Result — the tail is resolved below unity at both Reynolds numbers.

	D_{mean}	$D(q)/D_{\text{mean}}$, top-1%	$D(q)/D_{\text{mean}}$, top-0.1%
`isotropic8192` ($R_\lambda \approx 1300$)	0.117	0.91 [0.89, 0.92]	0.87 [0.83, 0.92]
`isotropic32768` ($R_\lambda = 2500$)	0.120	0.94 [0.93, 0.96]	0.93 [0.90, 0.97]

(Brackets are block-resampled 95% intervals.) At both Reynolds numbers the conditional ratio is **resolved below 1** at top-1% and top-0.1% — the interval excludes unity — so the most intense vorticity is *more* geometrically depleted than the bulk, the regularizing direction, now established (not merely a tendency) at R_λ up to 2500. The ratio decreases monotonically into the tail at both. This is the measurement the previous revision could not make; the $\sim 10^2$–$10^3\times$ larger, provenance-clean sample is what converts the order-unity per-snapshot scatter of the old data into stable, interval-bounded estimates.

The Reynolds comparison. Between the two datasets the tail depletion **weakens** as R_λ rises (the ratio moves toward 1): $0.91 \to 0.94$ at top-1% and $0.87 \to 0.93$ at top-0.1%. At top-1% the two intervals **do not overlap** ($[0.89, 0.92]$ vs $[0.93, 0.96]$), so the weakening is statistically resolved; at top-0.1% the direction is the same but the intervals marginally overlap, so there it is suggestive, not resolved. The direction — milder geometric depletion of the extreme tail at higher Reynolds number, a *mild de-depletion* — is exactly what Buaria, Pumir and co-workers report from independent high-R_λ DNS [BBP20, BPBY19, BP22].

Why we do not call this a Reynolds-number scaling law. Three reasons, stated plainly. (1) **Two points.** Two Reynolds numbers establish a difference, not a trend or a law. (2) **A confounded second axis.** `isotropic32768` is not only higher-R_λ but better resolved ($k_{\max}\eta \approx 2.8$ vs 1.4); we cannot fully separate Reynolds number from small-scale resolution with two datasets. We note the direction of the resolution bias is *conservative* here — under-resolution smooths the most intense gradients and pushes the conditional ratio *toward* 1, so the coarser `isotropic8192` would, if anything, read artificially *less* depleted, i.e. against the observed direction — but this argument bounds rather than removes the confound. (3) **The deepest tail is not resolved.** §3.2 shows the gradient stencils scatter 5–9% at top-0.01% (sign-flipping between datasets), so we restrict every claim to top-0.1% and shallower. The cross-check that the gate (point-gradient `getData` boxes) gave the *opposite* cross-R_λ ordering on a $\sim 300\times$ smaller sample — a disagreement we resolve in favour of the 10^8-point cutout measurement — is itself a reminder that the second-decimal comparison is at the edge of what this data supports.

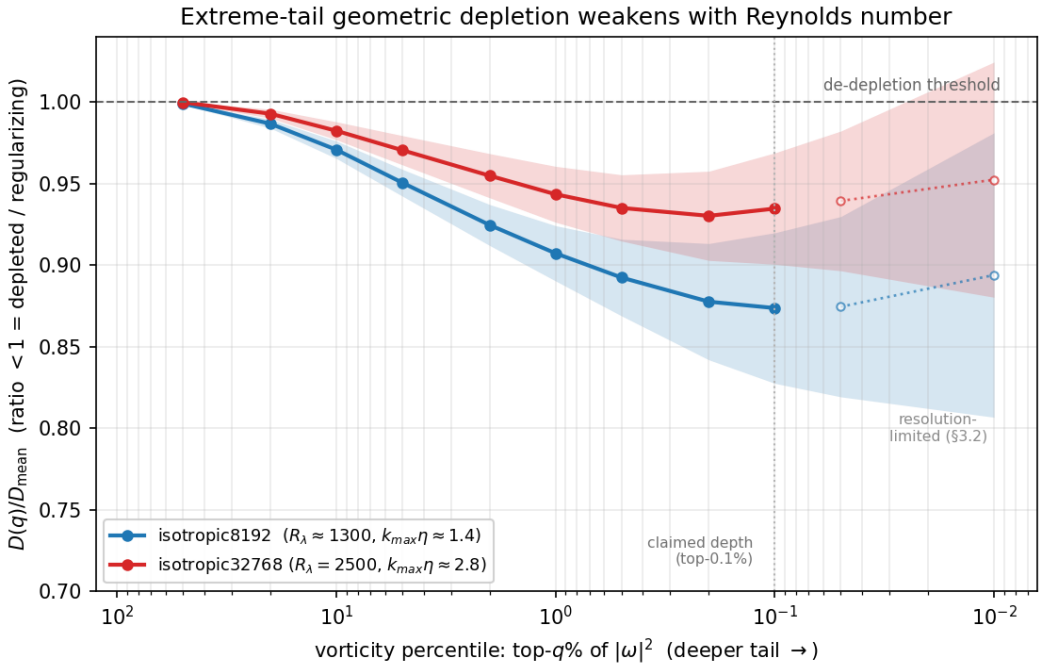


Figure 2: Conditional geometric-depletion ratio $D(q)/D_{\text{mean}}$ versus vorticity percentile for the two high-Reynolds JHTDB datasets, with block-resampled 95% bands ($K = 20$ contiguous blocks). Both curves lie **below** the de-depletion threshold (dashed, $= 1$) and decrease into the tail — the most intense vorticity is the most geometrically depleted (regularizing) — at both $R_\lambda \approx 1300$ and $R_\lambda = 2500$. The higher-Reynolds curve sits systematically **above** the lower (less depleted): the tail depletion *weakens* with Reynolds number, statistically resolved at top-1% (disjoint bands) and suggestive at top-0.1% (bands touch). Solid markers are the claimed depth (top-0.1% and shallower); faded dotted markers beyond it are resolution-limited (§3.2) and not claimed.

7. Model-problem corroboration: advection depletes blow-up (1D OSW family)

The 3D measurement is statistical and indirect — forced isotropic turbulence is a proxy, not a blow-up scenario (§10). To exhibit the *same* competition — nonlinear stretching versus depleting transport — in a setting where finite-time blow-up is **decidable**, we turn to the one-dimensional Okamoto–Sakajo–Wunsch (OSW) family [OSW08]:

$$\omega_t + a\, u\, \omega_x = u_x\, \omega, \qquad u_x = H[\omega],$$

with H the Hilbert transform and a the advection strength. At $a = 0$ this is the Constantin–Lax–Majda model [CLM85], which possesses an *exact* finite-time singularity; at $a = 1$ it is the De Gregorio model [DG90]. The rigorous picture, which we take care to state, is regularity- and domain-dependent: for **smooth** data on the **circle** (periodic domain, our setting) the De Gregorio advection is *conjectured* (and numerically observed) to suppress blow-up, but **low-regularity (C^α) data blows up for *every* a** — Chen, Hou & Huang [CHH21] prove finite-time, self-similar singularities for C^α data at all a (on $\mathbb{R}$ and S^1), and Elgindi & Jeong [EJ20] establish C^α blow-up and isolate the advection-vs-stretching competition. (On the line $\mathbb{R}$, [CHH21] also construct *smooth* compactly-supported blow-up for $a = 1$; the smooth-data regularity conjecture is specific to the periodic domain.) The coefficient a thus tunes the strength of the depleting transport for fixed (smooth) data — the 1D analogue of the geometric suppression of $\omega \cdot S\omega$ that, in 3D, arises from vorticity's alignment with the intermediate strain eigenvector.

We integrate the family by a Fourier pseudospectral method (2/3 dealiasing, CFL-adaptive RK4, $N = 4096$) from smooth data $\omega_0 = \sin x$, declaring blow-up when $\|\omega\|_\infty$ amplifies by 2×10^4. The solver reproduces the known endpoints exactly: $a = 0$ blows up at $t \approx 2.01$, while $a = 1$ stays flat — $\|\omega\|_\infty$ never exceeds its initial value, the advection cancelling the stretching outright. Scanning a and fitting the blow-up time to $t_{\mathrm{blow}}(a) = t_0 + C\,(a^* - a)^{-p}$ reveals a **depletion threshold** at which the singularity time *diverges*:

a	0.0	0.4	0.6	0.7	0.8	0.86	0.90	0.92	0.94	0.96
t_{blow}	2.01	2.79	3.63	4.39	5.87	7.81	10.50	14.22	29.85	— (regular to T=40)

The blow-up time **diverges steeply** as a increases, bracketing the critical strength $a^* \in (0.94,\, 0.96)$: at $a = 0.94$ the singularity forms at $t \approx 29.9$, while $a = 0.96$ remains bounded (amplification $\times 2.1$) under integration to $T = 40$ (2×10^5 steps). A power-law fit $t_{\mathrm{blow}}(a) = C\,(a^* - a)^{-p}$ over the blow-up points returns $a^* \approx 0.95$, $p \approx 0.5$. We stress a methodological point that is *itself* the heart of the open De Gregorio regularity question: because t_{blow} diverges so steeply near a^*, a fixed-time "survival" test **systematically misclassifies late-forming singularities as regular** — extending the integration window successively converted $a = 0.90$, 0.92, and 0.94 from apparently-regular to blow-up. The threshold must therefore be read from the divergence law (or a growth-rate criterion), not from survival to a finite T. We stress that $a^* \approx 0.95$ is the threshold **for this specific smooth datum ($\omega_0 = \sin x$) under finite-time integration**, not a universal critical advection strength: consistent with the rigorous result of Chen–Hou–Huang [CHH21] that C^α data blows up for *all* a, a separate low-regularity probe (cusp data $\sim |\sin x|^\alpha$) did exhibit blow-up at $a = 1$, but its

singular time collapsed toward zero under grid refinement — indicating the raw point-cusp is too singular to provide a clean finite-time test, and that a faithful reproduction of [CHH21] requires their specially-structured data rather than a generic cusp. The smooth-data result, by contrast, is sharp and reproducible. This is the mechanism of the 3D depletion story made decidable for smooth data: strengthening the transport that, in 3D, manifests as intermediate-eigenvector alignment, *continuously delays and then removes* the finite-time blow-up. The open 3D question (§5–6) is whether the analogous geometric depletion survives into the extreme tail at high Reynolds number (§6).

8. Discussion

At $R_\lambda \approx 433$ the *mean* depletion is firmly established and the tail point estimates indicate a regularizing tendency (ratio ≈ 0.77–0.81 in the large sample, decreasing into the tail), though the extreme top-0.1% ratio there is statistically marginal — the resampleable bootstrap straddles unity (§5). At the higher Reynolds numbers, where the §6 high-statistics measurement provides 10^2–$10^3\times$ more points, the picture sharpens into a resolved result: at $R_\lambda \approx 1300$ and $R_\lambda = 2500$ the extreme-tail ratio is **below unity with block-resampled intervals that exclude 1** at top-1% and top-0.1% — the geometric depletion of the most intense vorticity is real and persists to $R_\lambda = 2500$. Comparing the two, the depletion **weakens** with Reynolds number (resolved at top-1%, suggestive at top-0.1%), the mild de-depletion direction reported by Buaria–Pumir. This is the singularity-relevant direction the supercriticality barrier identifies as decisive: if the geometric protection erodes as R_λ grows, it erodes precisely where blow-up would be seeded. We are careful not to over-read it — two Reynolds numbers are a comparison, not a law; resolution co-varies with R_λ across the two datasets; and the deepest tail (top-0.01%) is not gradient-resolved (§3.2, §6). And none of this bears on blow-up directly: these are statistically steady forced flows, not a putative self-similar collapse; the measurement quantifies a geometric *trend*, not a singularity.

9. Completed measurements and what remains

The high-statistics program of the prior revision has been executed (a standard JHTDB authorization token lifts the 4096-point testing cap):

1. **High-statistics $R_\lambda \approx 1300$ tail — done.** Per-snapshot ($R_\lambda \approx 1300$ only) contiguous 128^3 cutouts of `isotropic8192`, 7.7×10^7 interior points; tail resolved below unity at top-1% and top-0.1% with block-resampled intervals (§6). As anticipated, the effective sample is far below the raw count (cutout points are spatially correlated), which is why we bound the tail by block (not point) resampling; the top-0.01%–0.001% are *not* claimed.
2. **The exascale DNS — done.** Same diagnostic on `isotropic32768` ($R_\lambda = 2500$, $k_{\max}\eta \approx 2.8$), 1.6×10^8 points: a second, higher-Reynolds point on the $D_{\text{tail}}/D_{\text{mean}}$ curve (§6).
3. **Differentiation-order cross-check — done.** The full stencil ladder (`fd4/fd6/fd8/m2q8`) at $\sim 5 \times 10^5$ identical grid points on both datasets confirms order-convergence through top-0.1% (D_{mean} ratio 1.00; top-0.1% agreement $< 1\%$) and locates the resolution frontier at top-0.01% (5–9% scatter, sign-flipping) (§3.2).

What remains, and would convert the two-point comparison into a trend:

4. **A third Reynolds number** between 433 and 2500 at matched resolution, to test monotonicity and begin a genuine $D_{\text{tail}}/D_{\text{mean}}(R_\lambda)$ curve, and ideally a resolution-matched pair to break the R_λ–$k_{\max}\eta$ degeneracy of §6.

5. **The deepest tail.** Resolving top-0.01%–0.001% — where any *crossing* of 1 (true de-depletion) would live — needs both more effective-independent samples ($\mathcal{O}(10^8$–$10^9)$) and a finer gradient than on-grid 4th-order FD provides at that depth.
6. **Conditioning.** Resolve D jointly on $|\omega|$ and $|S|$, and on the local strain-state (R–Q invariants), to identify *which* extreme events de-deplete.

10. Limitations

(i) Forced statistically-steady isotropic turbulence is a proxy for the regularity question, not a blow-up scenario. (ii) The high-Reynolds results resolve the top-1% and top-0.1% but **not** the top-0.01%–0.001%, where any true *crossing* of unity would live: the gradient stencils scatter 5–9% there (§3.2) and the effective independent sample is insufficient (§6, §9). (iii) The Reynolds comparison rests on **two** datasets that differ in small-scale resolution as well as R_λ ($k_{\max}\eta \approx 1.4$ vs 2.8); we therefore claim a two-point difference, not a scaling law, and note (§6) the resolution bias runs conservative to the observed direction. (iv) Block resampling preserves within-cutout correlation but $K = 20$ blocks give a coarse interval; a larger block count and additional independent snapshots would tighten it. (v) JHTDB and in-house Taylor–Green differ in forcing/decay; absolute D values are not directly comparable across them, so we rely on the *bulk-vs-tail ratio* as the within-dataset discriminator.

11. Conclusion

We have built and validated an apparatus that measures the percentile-resolved geometric depletion of enstrophy production across Reynolds number, reproduced the canonical alignment, and firmly established the mean depletion ($D \approx 0.12$). At $R_\lambda \approx 433$ the tail point estimates indicate a regularizing tendency (ratio $\to$ 0.77–0.81), though the resampleable bootstrap straddles unity at the top-0.1%. At $R_\lambda \approx 1300$ and $R_\lambda = 2500$, a high-statistics, per-snapshot, single-R_λ measurement (7.7×10^7 and 1.6×10^8 interior points) **resolves the extreme tail below unity** at top-1% and top-0.1% — block-resampled intervals exclude 1 — establishing that the most intense vorticity is more geometrically depleted than the bulk, and that this regularizing depletion persists to $R_\lambda = 2500$. Across the two Reynolds numbers the depletion **weakens** (mild de-depletion, in the Buaria–Pumir direction): resolved at top-1% (non-overlapping intervals), suggestive at top-0.1%. We deliberately frame this as a two-point comparison, not a Reynolds scaling law — the two datasets co-vary in resolution, and the deepest tail (top-0.01%), where a true crossing of unity would live, is not yet gradient-resolved. As an independent, *decidable* corroboration of the depletion *mechanism*, §7 exhibits the same transport-depletes-stretching competition in the one-dimensional Okamoto–Sakajo–Wunsch model family, where we locate a sharp blow-up/regularity threshold. The contributions of this work are thus the validated apparatus, the resolved extreme-tail depletion at $R_\lambda \leq 2500$ with its cautiously-stated Reynolds comparison, the $R_\lambda \approx 433$ baseline, the 1D-model corroboration, and a reproducible, self-audited account that holds each claim to the depth the data support.

12. Methodology note: an AI-augmented, consumer-hardware research apparatus

This work was produced by a small human-led, agentic-AI research process, and we report the methodology in the same spirit of disclosure as the results, because *how* it was done is part of what it demonstrates.

The apparatus. A human investigator (C.R.) set direction, posed questions, and served as the accountable validation layer; an agentic large-language-model system did the bulk of the literature mapping, code authoring, numerical experimentation, and drafting, under continuous human review. Two ingredients are worth naming. First, a **Topic-Keyed Hash Routing (TKHR)** memory substrate — a tamper-evident, hash-chained index that mirrors a soul/agent-memory pattern — was used to organize the relevant literature and the catalogue of "what does not work" for the Navier–Stokes regularity problem, and to surface candidate *cross-subdomain connections* (e.g. linking the 1D model-problem mechanism of §7 to the 3D geometric-depletion picture). Second, a capacity-aware LLM router fanned ideation and an adversarial literature-scrutiny pass across a mix of local and free/flat-rate cloud models. The proposed experiments were then implemented and run as ordinary, inspectable code.

Consumer hardware. The entire effort ran on commodity equipment: a single workstation (one RTX 5090 and one RTX 5070), the public Johns Hopkins Turbulence Database accessed through a free SciServer Compute container, and free or flat-rate LLM tiers. The in-house pseudospectral DNS, the JHTDB cutout pipeline, the 1D model-problem solver (§7), and the literature scrutiny were all feasible at this scale. What was *not* feasible at this scale is equally informative: the definitive extreme-tail measurement (§6, §9) exceeded the statistics our access tier could provide, and we say so rather than paper over it.

The integrity layer is the point. During this study the AI system produced, at three separate stages, confident and internally coherent conclusions about the high-R_λ extreme tail that did not survive scrutiny: (1) an apparent Reynolds trend that was an artifact of pooling mixed-R_λ snapshots; (2) a "converged" tail that was rare-event under-sampling at $\sim 5 \times 10^5$ points; and (3), in this revision, a small-sample point-gradient ("gate") comparison that put the cross-R_λ ordering the *wrong way*, reversed only when the 10^8-point cutout measurement and a block-resampled error bar were brought in. A fourth catch was external-source-shaped: a web search confidently asserted the two datasets were at *comparable* Reynolds number, which would have voided the entire comparison — the primary simulation paper [Yeung32768] shows `isotropic32768` is at $R_\lambda = 2500$, twice `isotropic8192`. Each was caught — by a human-prompted provenance audit, an error-bar discipline that respects sample correlation, and verification against primary sources rather than an LLM's paraphrase — before it could stand. We present this as the central methodological result: an AI research collaborator moves fast and finds real structure, but *validation* — deciding what is true enough to claim, and to what depth — must sit with an accountable human and an apparatus that distrusts its own first answer. The final status of this paper (a firmly-established mean depletion; a tail depletion now *resolved* below unity at $R_\lambda \leq 2500$; a Reynolds comparison stated as a two-point difference with explicit resolution and depth caveats; an unresolved deepest tail) is itself the product of that layer working.

On time-to-findings, and on disciplinary reach. This work was carried out over a span of days, by a human investigator with a background in engineering and security — *not* in turbulence or fluid dynamics. From a standing start the apparatus reproduced canonical vortex-stretching results, stood up a working GPU DNS, ingested and analysed state-of-the-art database turbulence, located the sharp threshold of a 1D model problem, and produced this manuscript with its literature scrutiny and self-audit — on commodity hardware. The point is not that the result is large (it is deliberately modest and hedged); it is that an agentic AI research partner can compress the *time to a checkable, honestly-bounded research artifact*, and can lower the disciplinary barrier to entering an unfamiliar field, provided an accountable human holds the validation layer. This mirrors the broader 2024–2026 evidence that agentic systems accelerate the *workflow* of science — literature synthesis,

hypothesis generation, code authoring, and closed-loop analysis — with reported compressions from months to days and hours to minutes in some pipelines, and with experimentally validated outputs in others (e.g. multi-agent "AI co-scientist" systems) [GoogleAICoSci25]. Our contribution to that conversation is methodological and cautionary in equal measure: the *speed* is real, and so is the failure mode (confident, well-formed, wrong intermediate conclusions); the two are reconciled only by a disciplined human-in-the-loop validation layer of exactly the kind we document here. We disclose AI use explicitly, in keeping with emerging journal norms on AI-assisted research and authorship [NatureAI23].

Author contributions and AI transparency statement

C.R. (Cristian Ruvalcaba) conceived and directed the study, made all substantive judgments, validated every claim, and is solely accountable for the contents. The **Agentic AI Research Team** (a large-language-model system operated by Saluca Labs) performed literature search and synthesis, authored the numerical codes (`dns_*`, `osw_*`, `sciserver_cutout.py`, analysis and bootstrap scripts), executed the computational experiments, drafted the manuscript, and ran the adversarial literature scrutiny — all under human direction and review. Listing the AI system as a co-contributor reflects the reality of how the work was done; it does not transfer scientific responsibility, which remains with the human author. No claim in this paper should be treated as established solely because an AI produced it; the human-validated, reproducible artifacts (code, data, and this disclosure) are the basis for trust.

Data and code availability

DNS solver, JHTDB pipeline (`sciserver_cutout.py`), the 1D OSW model solver (`osw_1d.py`, `osw_frontier.py`), analysis and bootstrap scripts, and figure code are archived with this work. JHTDB datasets are public [JHTDB08]; the geometric-depletion statistic and the per-snapshot audit of §6 are reproducible from them.

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